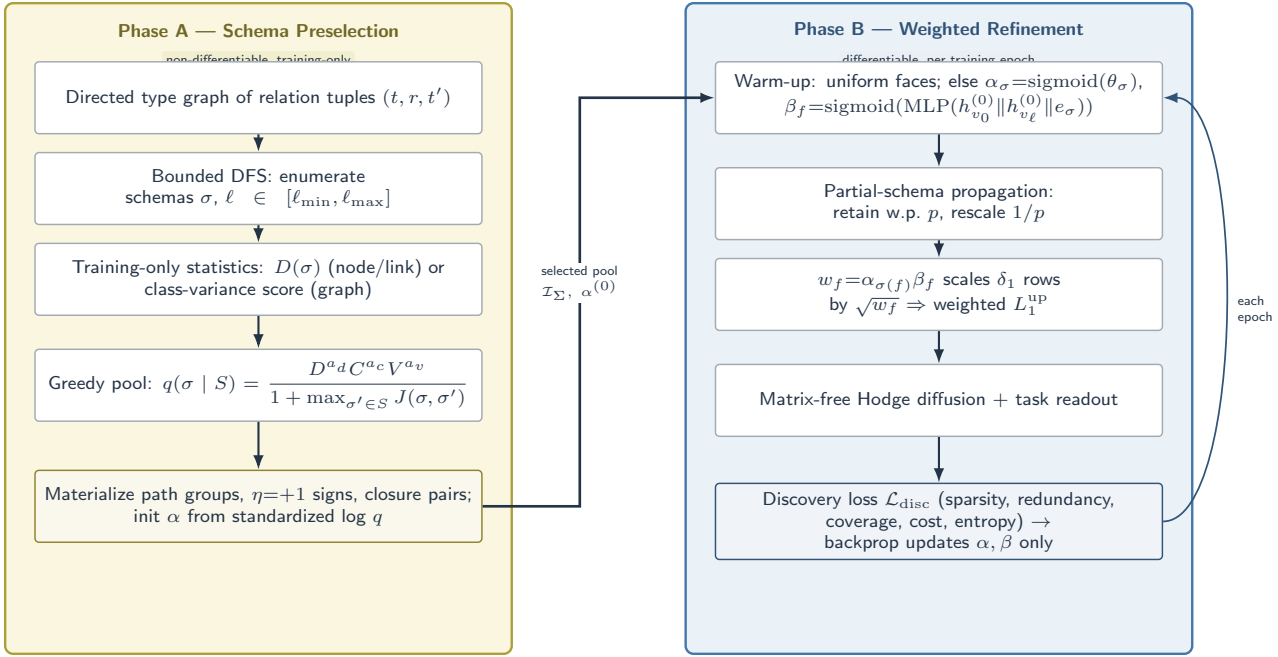

Phase A — Schema Preselection
non-differentiable, training-only
Directed type graph of relation tuples (t, r, t')
Bounded DFS: enumerate schemas σ, $\ell \in [\ell_{\min}, \ell_{\max}]$
Training-only statistics: $D(\sigma)$ (node/link) or class-variance score (graph)
Greedy pool: $q(\sigma \mid S) = \frac{D^{a_d} C^{a_c} V^{a_v}}{1 + \max_{\sigma' \in S} J(\sigma, \sigma')}$
Materialize path groups, η=+1 signs, closure pairs; init α from standardized log q
selected pool $\mathcal{I}_\Sigma$, $\alpha^{(0)}$
Phase B — Weighted Refinement
differentiable, per training epoch
Warm-up: uniform faces; else α_σ=sigmoid(θ_σ), β_f=sigmoid$(\mathrm{MLP}(h_{v_0}^{(0)} \| h_{v_\ell}^{(0)} \| e_\sigma))$
Partial-schema propagation: retain w.p. p, rescale $1/p$
w_f=$\alpha_{\sigma(f)}\beta_f$ scales δ_1 rows by $\sqrt{w_f}$ ⇒ weighted L_1^{up}
Matrix-free Hodge diffusion + task readout
Discovery loss $\mathcal{L}_{\mathrm{disc}}$ (sparsity, redundancy, coverage, cost, entropy) → backprop updates α, β only
each epoch